**SEAGLASS FOUNDRY**

*Forged For Speed. Refined by Craft.*

# ForgeData™

## *The Data-Intelligence Layer of the Forge Suite*

ForgeData™ decides which geospatial data to use and how to get it. A real workload pulls elevation, imagery, hydrography and point clouds from a dozen incompatible places; ForgeData answers, on every routing request, which datasets cover the area, which are at the right resolution, which is cheapest to read, and whether the suite has already computed the result. It is bundled with the Forge suite, so ForgeGIS™, ForgeMind™ and the suite's map views stay focused on what each does best.

**14 → 1**

**source types, one ranked answer**

*PostGIS, STAC, ArcGIS, OGC WCS and WFS, USGS 3DEP, OpenTopography, NOAA, local folders and more*

**One file**

**the whole catalog, portable**

*datasets and the recipes that refresh them in one GeoPackage — safe to copy, ship, or keep offline*

**Stated**

**units and datum on every elevation answer**

*new in 1.2.6: vertical units, datum and where they came from — reported as stored, never guessed*

### What it is

ForgeData sits beneath the suite and owns the cross-cutting problem of data discovery and access. It keeps one catalog — what you can query and the recipes that refresh it — and routes every request to the best-fit dataset across local files and remote services alike, with no hardcoded paths and no per-project glue.

### Why it matters

Most platforms make the customer build the routing layer: bespoke catalog tables, hand-maintained config, hardcoded paths nobody wants to own. ForgeData ships that layer in the box, so the suite works against a realistic, scattered data estate on day one rather than after an integration project.

### How it's built differently

The catalog is the configuration — source recipes live in the same file as the inventory, so nothing drifts, and keys stay in the environment, never the file. Routing is fidelity-first: adequate coverage, then the finest resolution, then extent-fit, with access cost only breaking the tie, and remote costs calibrate themselves on measured latency.

**Right answers, checked.** Release 1.2.6 is a correctness release: projected DEMs and LiDAR in feet now read correctly, every distance and area is measured on the WGS84 ellipsoid, and core queries are checked against GDAL.

**In the Forge suite.** ForgeMind™ and Seaglass Globe™ run ForgeData as an MCP server and ask it for data; ForgeGIS Studio™ embeds it as a library; all of them read the same catalogs. ForgeData reads through ForgeGIS™ in-process, so extraction and elevation queries make no network hop, and the underlying sources stay where they are.



CONTINUED ON REVERSE →

## WHO FORGEDATA IS FOR

### *Suite buyers who want it to just work against real data*

When a ForgeMind workflow asks for a viewshed or a slope analysis, the agent does not need to know one dataset is a local SRTM tile and another a remote COG on S3. It asks once and gets back a ranked, costed list — best-fit correct dataset first — and when nothing covers the area, it is told what is missing and which registered source could fetch it.

### *Teams with a mixed data estate*

PostGIS, STAC, ArcGIS REST, Overpass, OGC WCS and WFS, OpenTopography, USGS 3DEP, NOAA, Copernicus, Mapzen Terrarium, TIGER/Line roads and addresses, cloud COGs and plain local folders all answer through one query interface. The environment can be a mix; the caller sees one ranked answer.

### *Workstations shared by a team*

Since 1.2.5, the Seaglass Control Center™ gives each person their own catalog and, on an install for everyone, a read-only catalog for the whole machine beside it. An Administrator publishes a folder once and every account can find it. Seaglass Globe™ scenarios can be registered alongside the data they sit on.

## BUILT TO BE TRUSTED

> **Pre-computed work is remembered.** When ForgeGIS computes a slope raster over an area, the result registers back into the catalog as a derived product that records the operation and parameters that made it. The next request that needs the same thing finds it instead of recomputing it.

> **Refuses, never fabricates.** Failures are structured, machine-readable reasons — area out of coverage, bad bounding box, unsupported read — and since 1.2.6 every reply with no data says why, never the nearest wrong tile. Replies stay bounded: dataset lists page, and large geometry stays out of the reply and can be written to a sandboxed file on request.

> **Healthchecks built in.** Presence verification and a diagnostic doctor command find missing files, orphaned bytes and corrupted content, so operators see the catalog's real state without writing their own scripts.

## Get the full Technical Brief

The Technical Brief covers the catalog, the cost model, the operation registry, every source type, the MCP surface and deployment. ForgeData is bundled with the Forge suite and is not sold separately. The Forge suite is commercially licensed; evaluation builds are available to qualified teams.

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